

MTE 203 – Advanced Calculus

Week 6 Homework

Tangent Lines and Tangent Planes (S.12.9.)

Problem 1: [12.9, Prob. 19]

Find equations for the tangent line to the curve

$$x = t^2, \quad y = t, \quad z = \sqrt{t + t^4}$$

at the point $(1, 1, \sqrt{2})$

Problem 2: [12.9, Prob. 25]

Find an equation for the tangent plane to the surface

$$x = y \sin\left(\frac{\pi z}{2}\right)$$

at the point $(-1, -1, 1)$

Problem 3: [12.9, Prob. 31]

Find the directional derivative for the function

$$f(x, y, z) = xyz + xy + xz + yz$$

at $(1, -2, 5)$ perpendicular to the surface

$$z = x^2 + y^2$$

Problem 4: [12.9, Prob. 39]

Show that the sum of the intercepts on the x -, y -, and z -axes of the tangent plane to the surface $\sqrt{x} + \sqrt{y} + \sqrt{z} = \sqrt{a}$ at any point is a .

More practice problems¹

The answers to these problems, along with all even-numbered exercises, are provided at the back of the textbook. Detailed solutions can be found in Trim's Student Solution Manual. If you don't own a copy, a reserved edition is available at the Davis Centre.

Warm-Up Problems

1. S. 12.9, Probs. 2, 10, 22`

Extra Practice Problems

1. S. 12.9, Probs. 34, 36

Extra Challenge Problem

1. S. 12.9, Probs. 38

¹ You are strongly encouraged to try and solve the problems assigned in this homework on your own, before looking at the solutions. These problems are only the minimum necessary to master the material. You are also encouraged to do additional problems from the text for practice (some suggested **extra practice problems** have been listed here).